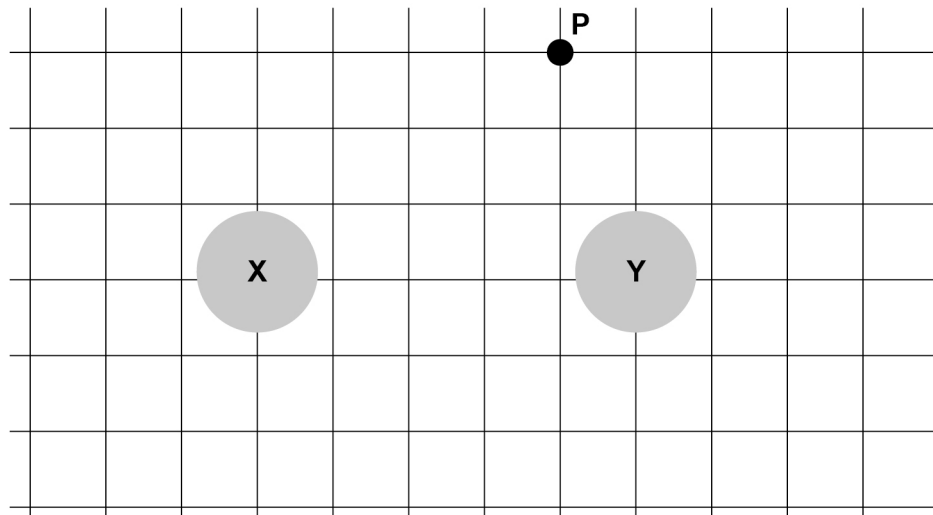


1 6

The scale diagram shows two positively charged spheres, **X** and **Y**.
The charge of **X** is five times the charge of **Y**.

The electric field strength at **P** due to **X** is E_X .
The electric field strength at **P** due to **Y** is E_Y .



drawn
to scale

What is true for E_X and E_Y ?

[1 mark]

A $E_X = 2E_Y$

☐

B $2E_X = E_Y$

☐

C $2E_X = 5E_Y$

☐

D $5E_X = 2E_Y$

☐

A positron is at a point in a uniform electric field where the electric field strength is 2.4 kV m^{-1} .

The acceleration of the positron is

[1 mark]

A $4.2 \times 10^{11} \text{ m s}^{-2}$ in the opposite direction to the electric field strength.

☐

B $4.2 \times 10^{11} \text{ m s}^{-2}$ in the same direction as the electric field strength.

☐

C $4.2 \times 10^{14} \text{ m s}^{-2}$ in the opposite direction to the electric field strength.

☐

D $4.2 \times 10^{14} \text{ m s}^{-2}$ in the same direction as the electric field strength.

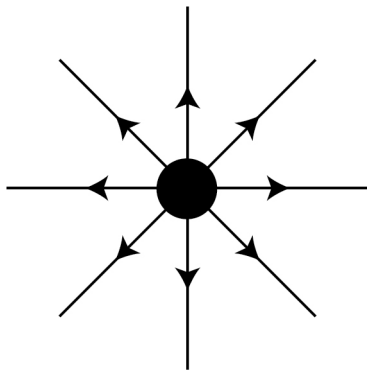
☐

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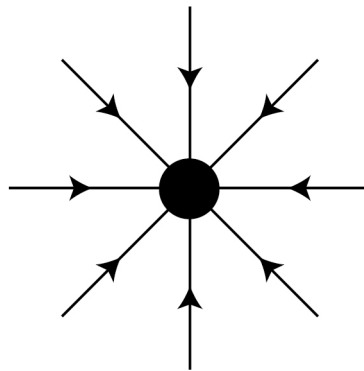


Do not write

The diagrams show two patterns of field lines, **X** and **Y**, around a particle.



X



Y

Which row is correct when the particle is a proton?

[1 mark]

	Gravitational field	Electric field	
A	X	X	☐
B	X	Y	☐
C	Y	X	☐
D	Y	Y	☐

Turn over for the next question

Turn over ►



Do not write Do not write

An electron enters a uniform electric field between two parallel plates.

The initial velocity of the electron is parallel to the plates.



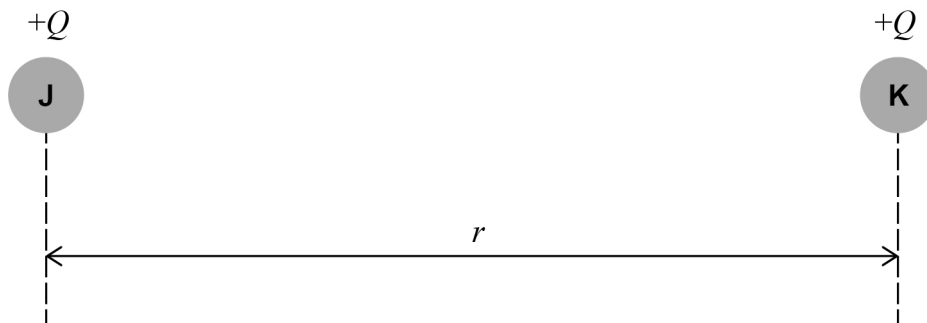
The acceleration of the electron is always

[1 mark]

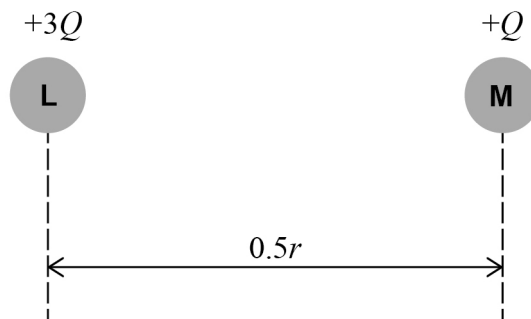
- A** perpendicular to the instantaneous velocity.
- B** perpendicular to the electric field strength.
- C** in the same direction as the electric field strength.
- D** in the opposite direction to the electric field strength.

☐☐☐☐

J and **K** are two point charges separated by a distance r .



L and **M** are two different point charges separated by a distance $0.5r$.



The force between **J** and **K** is F .

What is the force between **L** and **M**?

[1 mark]

A $\frac{3F}{2}$

☐

B $6F$

☐

C $12F$

☐

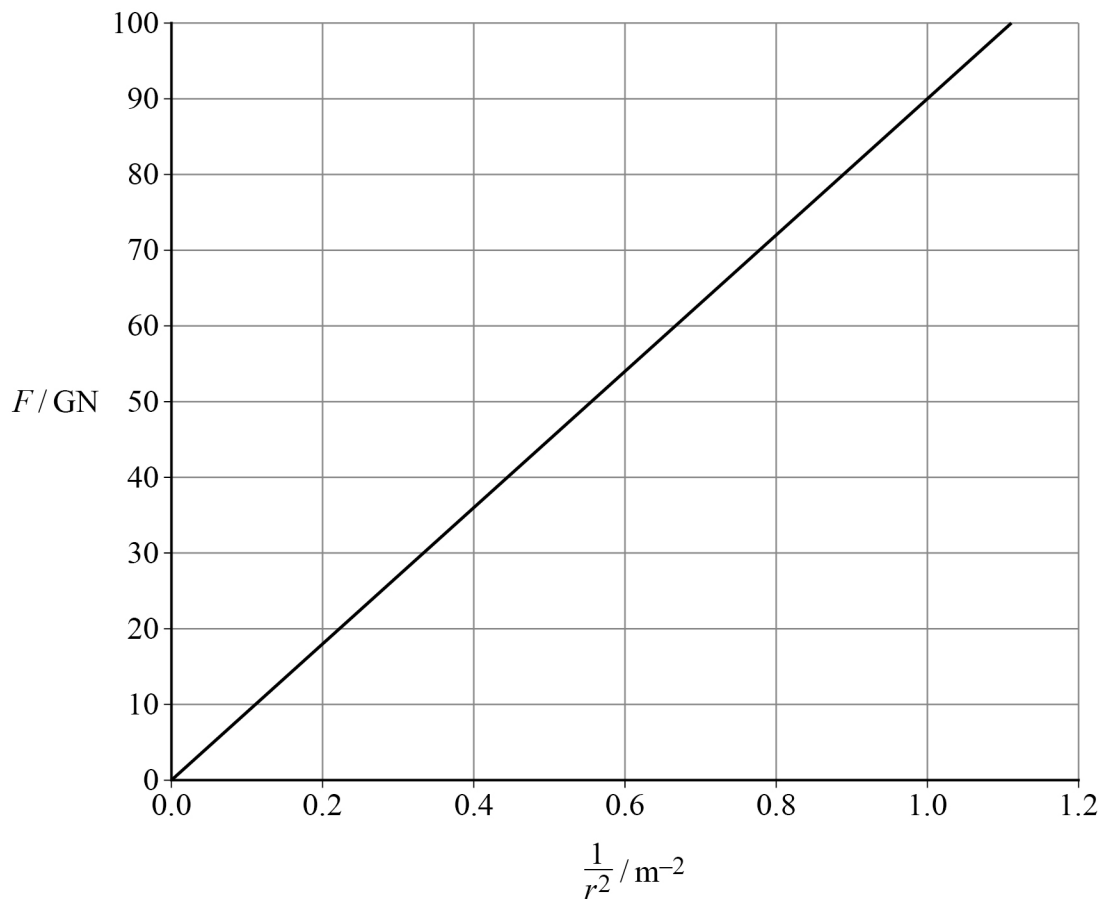
D $36F$

☐


Do not write

Two point charges experience a force F between them when separated by a distance r .

The graph shows how F varies with $\frac{1}{r^2}$.



One charge has a magnitude of 2.5 C.

What is the magnitude of the other charge?

[1 mark]

A 4 mC

☐

B 10 mC

☐

C 4 C

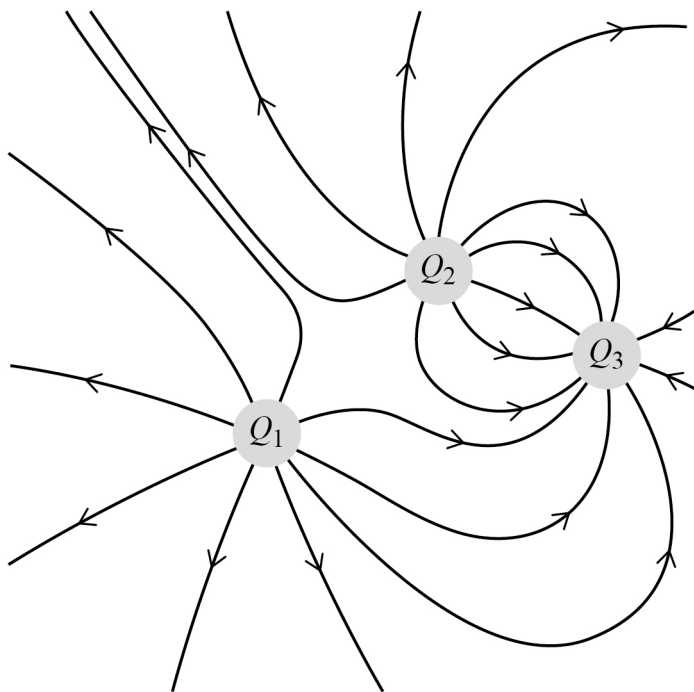
☐

D 10 C

☐


Don't waste

The diagram shows electric field lines around three charges Q_1 , Q_2 and Q_3 .



Which row gives the signs of the charges Q_1 , Q_2 and Q_3 ?

[1 mark]

	Q_1	Q_2	Q_3	
A	negative	negative	positive	☐
B	negative	positive	negative	☐
C	positive	negative	positive	☐
D	positive	positive	negative	☐



Don't waste

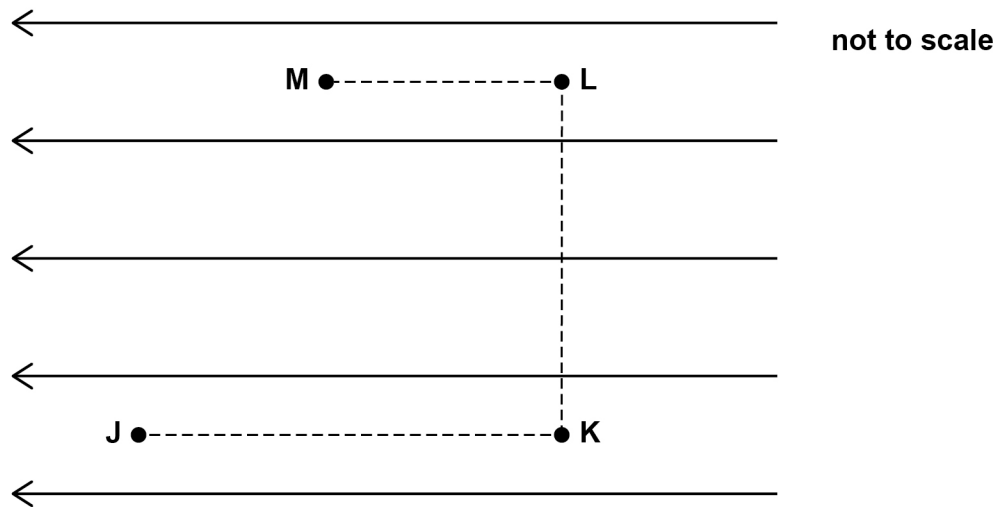
The diagram shows a uniform electric field strength E of 20 V m^{-1} .

A charge of $+8.0 \text{ mC}$ is moved along the path **JKLM**.

The distance **J** to **K** is 2.0 m .

The distance **K** to **L** is 1.0 m .

The distance **L** to **M** is 1.0 m .



What is the net work done moving the charge from **J** to **M**?

[1 mark]

A 160 mJ

☐

B 230 mJ

☐

C 480 mJ

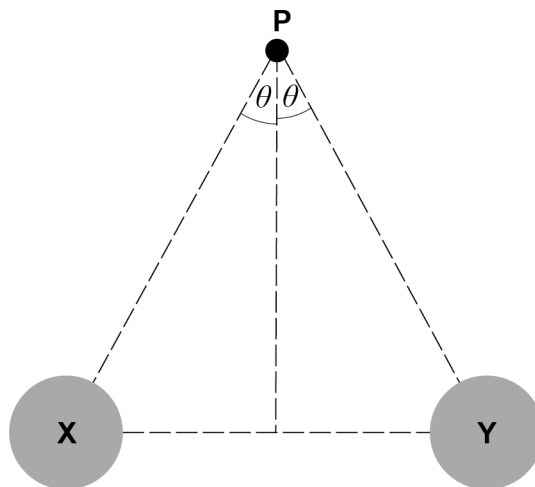
☐

D 640 mJ

☐


Do not write

P is a point equidistant from two charged particles **X** and **Y**.
The charge on **X** is $+Q$ and the charge on **Y** is $-2Q$.



The electric field strength at **P** due to **X** is E_X .

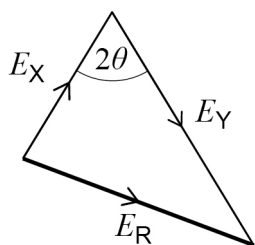
The electric field strength at **P** due to **Y** is E_Y .

Which diagram on page 33 shows how E_X and E_Y produce the resultant electric field strength E_R at **P**?

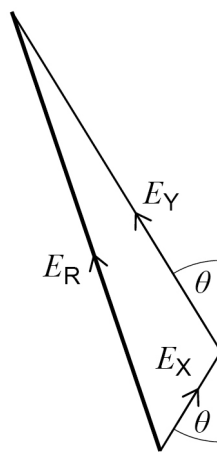
[1 mark]



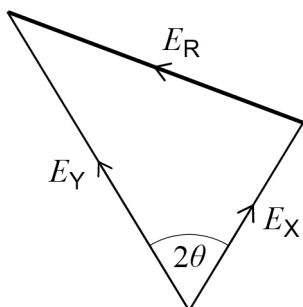
A



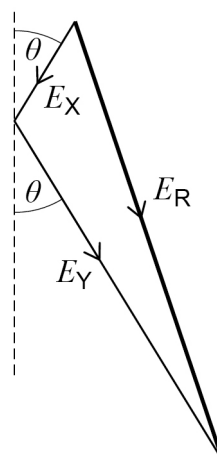
B



C



D



A

☐

B

☐

C

☐

D

☐


Do not write

1**3**

Two charged objects are placed at different positions in the same uniform electric field.

Which quantity must be the same for the two charged objects?

[1 mark]

A electric field strength

☐

B electric force

☐

C electric potential

☐

D electric potential energy

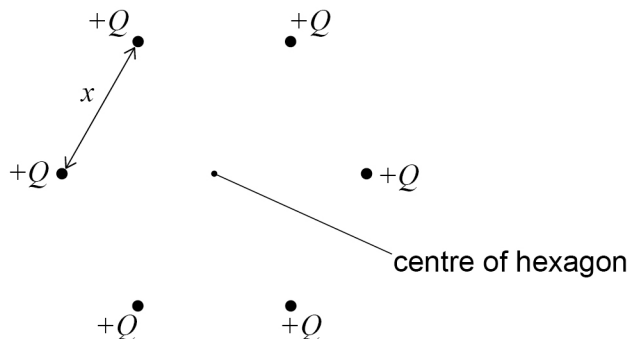
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4

Identical positively charged particles are fixed at the points of a regular hexagon of side length x . The magnitude of each charge is Q .



What are the magnitudes of the resultant electric field strength and the resultant electric potential at the centre of the hexagon?

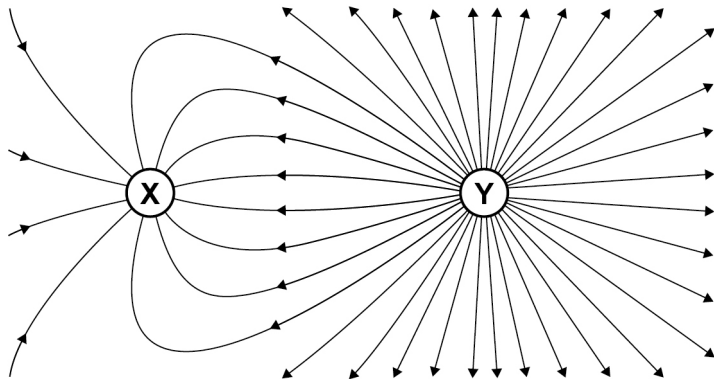
[1 mark]

	Resultant electric field strength	Resultant electric potential
A	zero	zero
B	zero	$\frac{6Q}{4\pi\epsilon_0 x}$
C	$\frac{6Q}{4\pi\epsilon_0 x^2}$	zero
D	$\frac{6Q}{4\pi\epsilon_0 x^2}$	$\frac{6Q}{4\pi\epsilon_0 x}$

☐
☐
☐
☐


do not write

The diagram shows a field around two objects **X** and **Y**.



Which is correct for this field?

[1 mark]

	Type of field	Relationship between X and Y	
A	electric field	magnitude of charge on X > magnitude of charge on Y	☐
B	electric field	magnitude of charge on Y > magnitude of charge on X	☐
C	gravitational field	mass of X > mass of Y	☐
D	gravitational field	mass of Y > mass of X	☐

Turn over for the next question



Do not write Do not write

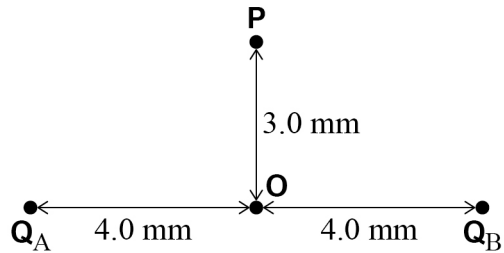
0 4

Figure 7 shows two identical negative point charges, Q_A and Q_B , in fixed positions 8.0 mm apart.

O is a point midway between the charges.

P is a point 3.0 mm away from **O** and equidistant from Q_A and Q_B .

Figure 7



0 4 . 1

The resultant electric field strength at **P** is 1400 N C^{-1} towards **O**.

Calculate the charge of Q_A .

[3 marks]

charge = _____ C



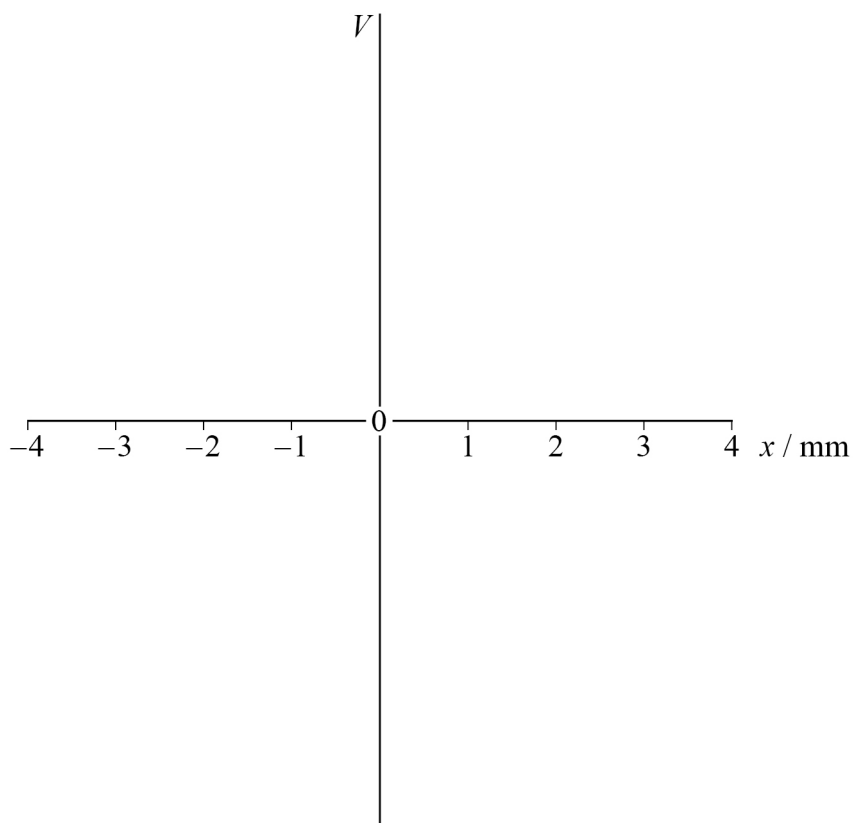
0 4 . 2

Sketch, on **Figure 8**, the variation of electric potential V with displacement x from **O** between **Q_A** and **Q_B**.

Calculations and numerical values for V are not required.

[2 marks]

Figure 8



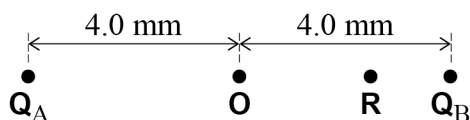
Question 4 continues on the next page

Turn over ►



Figure 9 shows a point **R** between **O** and **Q_B**.

Figure 9



Q_A and Q_B are in fixed positions.

A small mass **m** with negative charge q is released from rest at **R**.

Only electrostatic forces act on **m**.

After release, **m** oscillates about **O** along the line between Q_A and Q_B .

0 4 . 3

State and explain how the direction of the resultant force on **m** varies during oscillation.

[2 marks]

0 4 . 4

Describe the energy changes of **m** from its release until it reaches **O** for the first time.

[2 marks]

0 4 . 5

Deduce whether or not **m** performs simple harmonic motion.

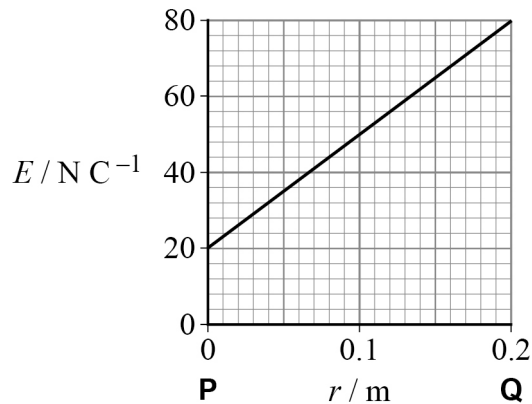
[1 mark]

Don't write

Points **P** and **Q** are 0.2 m apart in an electric field. The graph shows the variation of electric field strength E with position r between **P** and **Q**.

What is the potential difference between **P** and **Q**?

[1 mark]



A 6 V

☐

B 10 V

☐

C 60 V

☐

D 300 V

☐

Turn over ►



Do not write

Electrostatic precipitators are used to remove smoke particles from the gases released by coal-fired power stations.

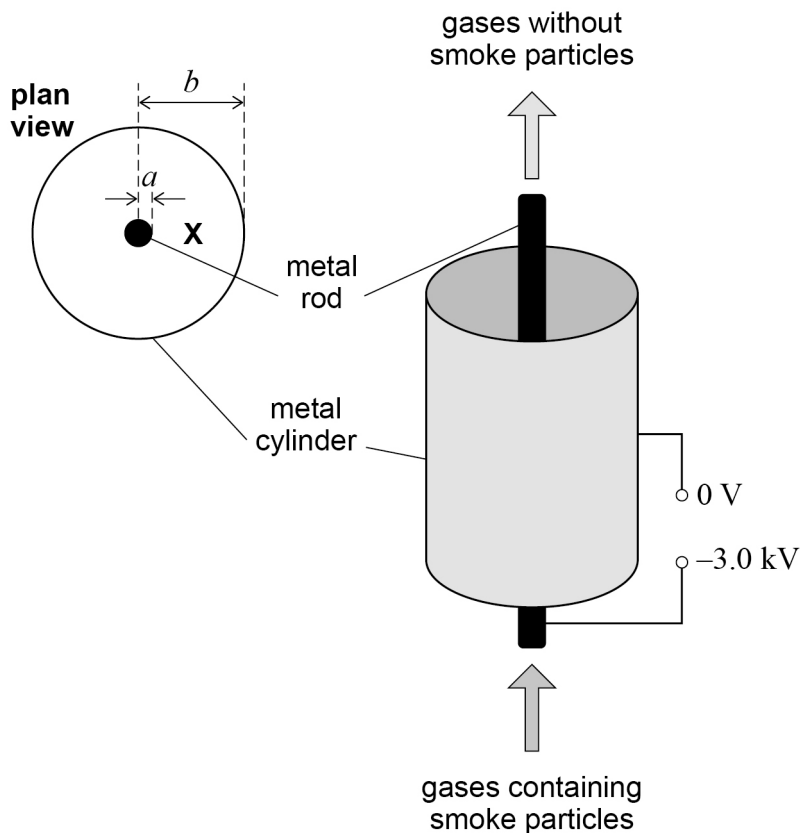
Figure 5 shows a simplified electrostatic precipitator.

A metal rod of radius a is placed along the vertical axis of a hollow metal cylinder of radius b .

Point **X** is halfway between the rod and the cylinder wall.

The rod and cylinder are connected to a high-voltage supply to create an electric field inside the cylinder. The electric field strength E is horizontal inside the cylinder.

Figure 5



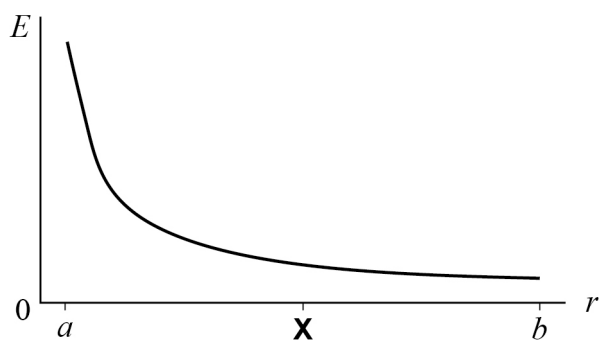
Gases containing smoke particles enter at the bottom of the cylinder. The smoke particles become charged and are attracted to the cylinder wall.

The smoke particles stick to the inside of the wall. The gases leave the top of the cylinder without the smoke particles.

Question 3 continues on the next page

Figure 6 shows the variation of the magnitude of E with radial distance r from the surface of the rod to the cylinder wall.

Figure 6



The rod is at an electric potential of -3.0 kV and the cylinder is at 0 V.

0 3 . 1

A student suggests that the potential at X is -1.5 kV.

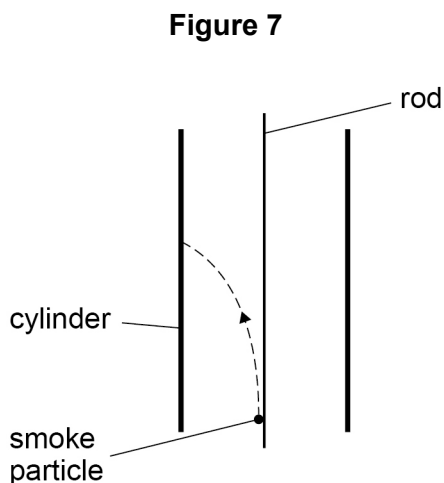
Discuss this suggestion.

[2 marks]



A smoke particle enters the cylinder near the rod. Initially it is uncharged and travels vertically with a velocity of 5.2 m s^{-1} .

Figure 7 shows the path of the smoke particle inside the cylinder.



The mass of the particle is $2.7 \times 10^{-17} \text{ kg}$.

The particle acquires a charge of $-3.2 \times 10^{-19} \text{ C}$ near the rod.

The particle experiences an initial horizontal acceleration of 91 m s^{-2} due to the electric field in the cylinder.

0 3 . 2 Determine the magnitude of E at the point where the particle acquires the charge.

[3 marks]

$$E = \underline{\hspace{10cm}} \text{ N C}^{-1}$$

Question 3 continues on the next page



The vertical component of the velocity of the particle remains constant at 5.2 m s^{-1} . Assume that all other forces on the particle are negligible except for the electrostatic force.

0 3 . 3

State and explain any change to the magnitude and to the direction of the acceleration of the particle as it moves along its path.
Refer to **Figure 6** in your answer.

[3 marks]

0 3 . 4

Calculate the speed of the particle as it reaches the cylinder wall.

[4 marks]

speed = _____ m s^{-1}



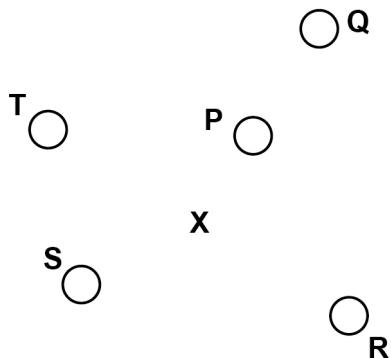
Don't write

1 2

Five charged objects are at points **P**, **Q**, **R**, **S** and **T**.

The objects are arranged so that the electric field strength at **X** is zero.

The object at **S** carries a positive charge.



The object at **S** is removed without affecting the other objects' positions or charge.

What is the direction of the resultant electric field strength at **X** due to the remaining four charged objects?

[1 mark]

A from **P** to **X**



B from **S** to **X**



C from **X** to **P**



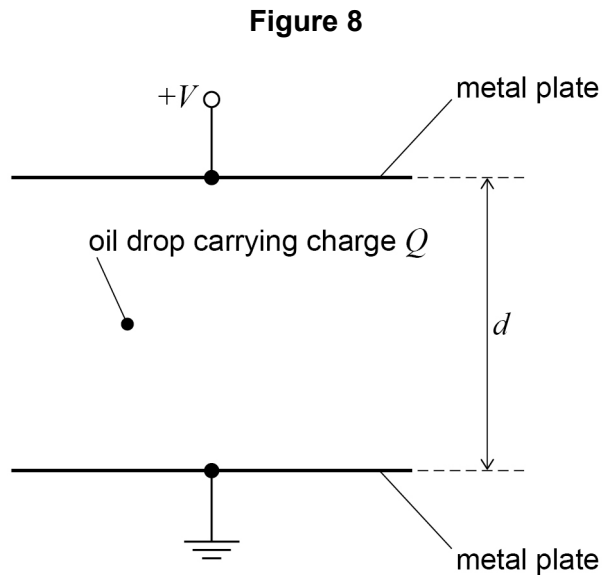
D from **X** to **S**



do not write

A spherical oil drop has a mass m , a radius r and carries a charge Q .

Figure 8 shows the oil drop between two horizontal metal plates. There is a potential difference V between the plates. The separation of the plates is d .



The oil drop is stationary because the forces acting on the oil drop are in equilibrium. Ignore any forces on the oil drop due to the air.

Show that $Q = -\frac{mgd}{V}$

[2 marks]

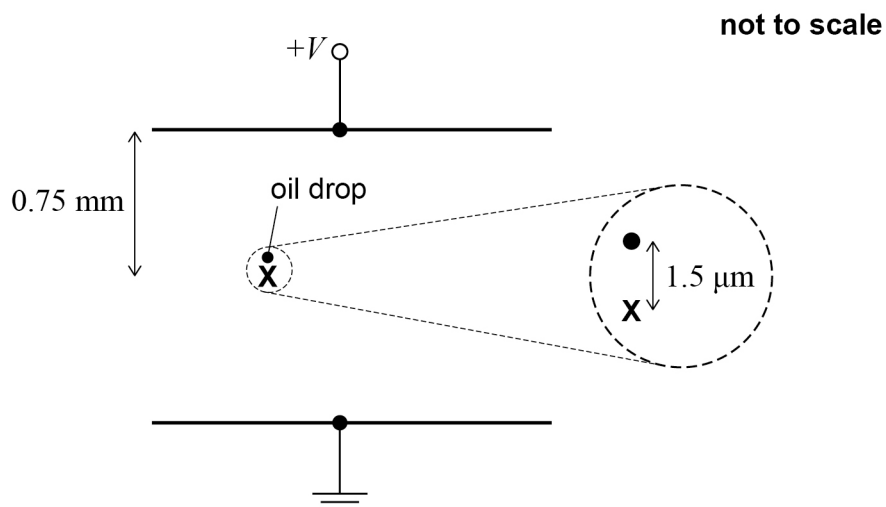
Question 4 continues on the next page

Turn over ►

0 4 . 2

Figure 9 shows point **X** that is 0.75 mm below the positive plate.
The centre of the oil drop is 1.5 μm vertically above **X**.

Figure 9



Determine the magnitude and direction of the resultant electric field strength at **X**.

$$V = 16 \text{ V}$$

$$d = 1.5 \text{ mm}$$

$$Q = -4.8 \times 10^{-19} \text{ C}$$

[5 marks]

magnitude of resultant electric field strength = _____ N C^{-1}

direction = _____



0 4 . 3

A change occurs to the situation shown in **Figure 8** so that the oil drop accelerates downwards.

Suggest **two** changes that can each cause this acceleration.

[2 marks]

1 _____

2 _____

9

Turn over for the next question

Turn over ►



Do not write

Q_1 and Q_2 are point charges of $+3.0 \mu\text{C}$ and $-3.0 \mu\text{C}$ respectively.

Q_1 and Q_2 are placed at different positions on an equipotential surface that has an electric potential of 25 kV.

The positions of Q_1 and Q_2 are exchanged.

What is the net work done in moving Q_1 and Q_2 to their new positions?

[1 mark]

A -0.075 J

☐

B 0

☐

C 0.075 J

☐

D 0.15 J

☐

Don't write

Coulomb's law can be written as

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$$

This equation is used to determine the force of one charged sphere on another charged sphere. Both charged spheres are in air.

What must be true?

[1 mark]

A r is the separation of the centres of the spheres.

☐

B The two charges carry the same type of charge.

☐

C There are no other charges nearby.

☐

D The relative permittivity of the air is added to ϵ_0 .

☐

[illegible]

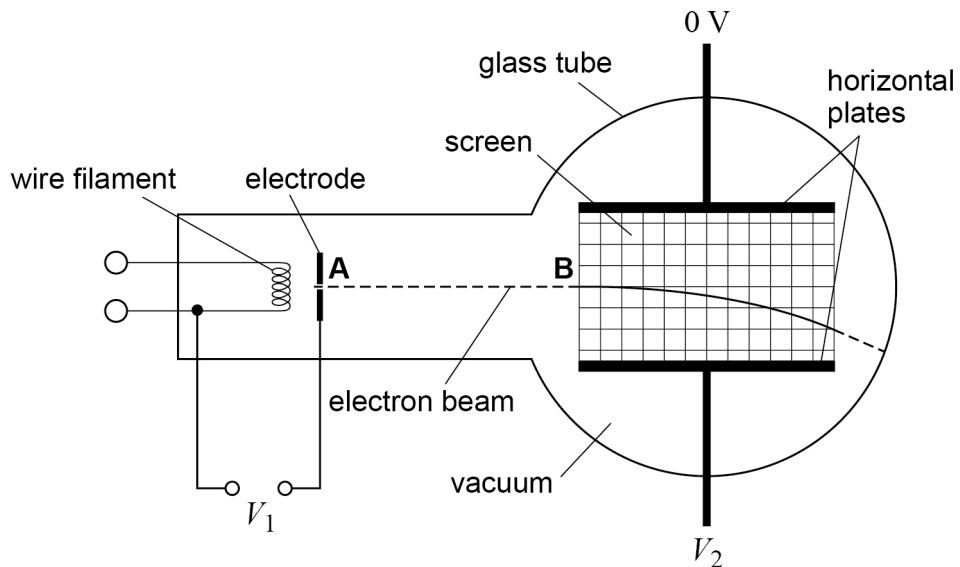
[illegible]

[illegible]

[illegible]

Figure 4 shows an arrangement that is used to investigate the properties of electrons.

Figure 4



Electrons are emitted from a wire filament with a negligible initial speed. The electrons are accelerated through a potential difference V_1 towards an electrode. The electrons that pass through the hole at **A** have a speed of $1.48 \times 10^7 \text{ m s}^{-1}$.

Calculate V_1 .

[2 marks]

$V_1 =$ _____ V



Figure 4 shows the electron beam entering a uniform electric field between two horizontal plates.

At **B**, the electrons are moving horizontally.

The uniform electric field causes the electrons to accelerate.

0 3 . 2

Describe this acceleration.

Explain your answer.

[3 marks]

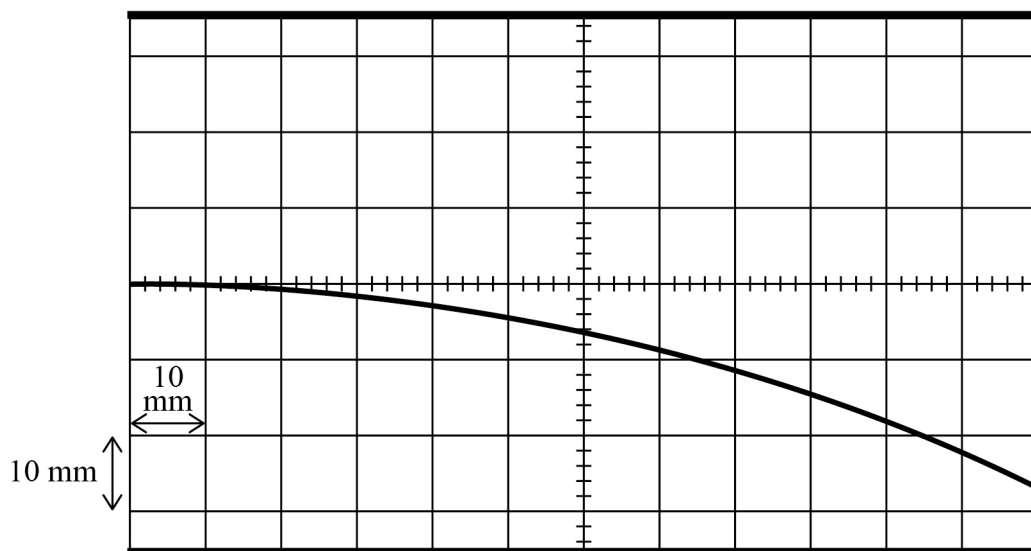
Question 3 continues on the next page



The curved line on **Figure 5** shows the path of electrons on the fluorescent screen between the horizontal plates.

Figure 5

to scale



Each square on the screen measures 10 mm by 10 mm.

0 3 . 3 The horizontal component of an electron's velocity is $1.48 \times 10^7 \text{ m s}^{-1}$.

Show that the time taken for the electron to cross the screen is about 8 ns.

[1 mark]



0 3 . 4 The potential difference across the plates is equal to V_2 .

Determine V_2 .

[4 marks]

$V_2 =$ _____ V

0 3 . 5 The apparatus is enclosed in a glass tube.
Initially there is no air in the glass tube.
A fault occurs and some air leaks into the tube.

Explain **two** effects on the curved line seen on the screen.

[2 marks]

1 _____

2 _____



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ANSWER IN THE SPACES PROVIDED**



Don't write

0	5
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This question is about a single proton as it moves through a linear accelerator.

In the linear accelerator, the proton is accelerated by a uniform electric field. The uniform electric field is created by a potential difference V that acts across a distance of 2.0 cm.

The electric field strength is 13.3 MN C^{-1} .

0	5	.	1
---	---	---	---

Calculate V .

Give your answer to an appropriate number of significant figures.

[3 marks]

$V =$ _____ V

0	5	.	2
---	---	---	---

Calculate the acceleration of the proton.

mass of proton = $1.67 \times 10^{-27} \text{ kg}$

[3 marks]

acceleration = _____ m s^{-2}

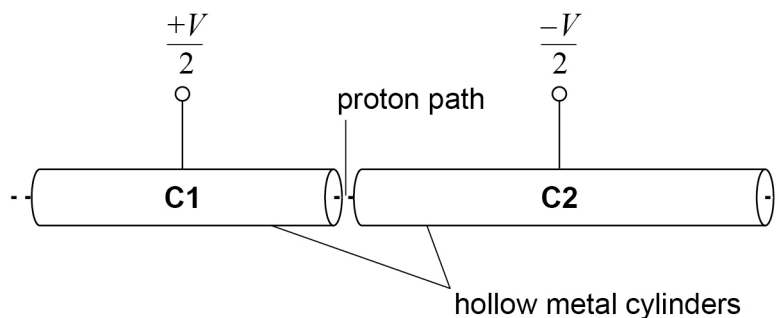


In the linear accelerator the proton travels through a series of hollow metal cylinders.

Figure 8 shows the first two hollow metal cylinders **C1** and **C2**.

V is applied between **C1** and **C2** to accelerate the proton as it moves from **C1** to **C2**.

Figure 8



The surface of **C1** is at a potential of $\frac{+V}{2}$ and the surface of **C2** is at a potential of $\frac{-V}{2}$.

The linear accelerator is inside a vacuum.

0 5 . 3 Explain why the acceleration of the proton is zero inside **C1** and inside **C2**.

[3 marks]

Question 5 continues on the next page

Turn over ►



0 5 . 4

The proton takes the same time to pass through each metal cylinder.

Explain why the length of **C2** is greater than the length of **C1**.

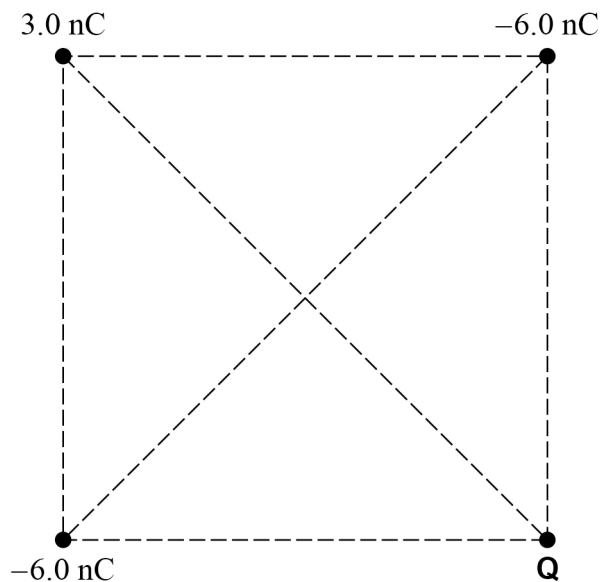
[2 marks]

11



Do not write

Four point charges are placed at the corners of a square as shown.



The electric potential is zero at the centre of the square.

What is the charge of Q ?

[1 mark]

A -9.0 nC ☐

B -3.0 nC ☐

C 3.0 nC ☐

D 9.0 nC ☐



Do not write

03.1

Figure 3 shows an isolated proton **P**.

Draw on **Figure 3** the electric field due to **P**.

[1 mark]

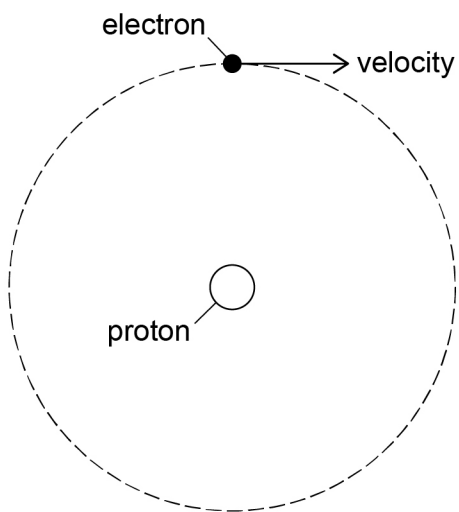
Figure 3

not to scale



In an early model of a hydrogen atom, the electron orbits the proton in a circular path, as shown in **Figure 4**.

Figure 4



not to scale

In this model, the only force acting on the electron is the electrostatic force exerted by the proton.

The distance between the centre of the electron and the centre of the proton is 5.29×10^{-11} m.



03.2

Determine the ratio $\frac{\text{electrostatic force}}{\text{gravitational force}}$ for the forces between the proton and the electron in **Figure 4**.

[3 marks]

$$\frac{\text{electrostatic force}}{\text{gravitational force}} = \underline{\hspace{10cm}}$$

03.3

Show that the kinetic energy of the electron is approximately $2 \times 10^{-18} \text{ J}$.

[3 marks]

Question 3 continues on the next page

Turn over ►



0 3 . 4

Calculate the energy required to completely separate the orbiting electron from the proton.

Assume that the electron has only kinetic energy and electric potential energy.

[3 marks]

energy = _____ J

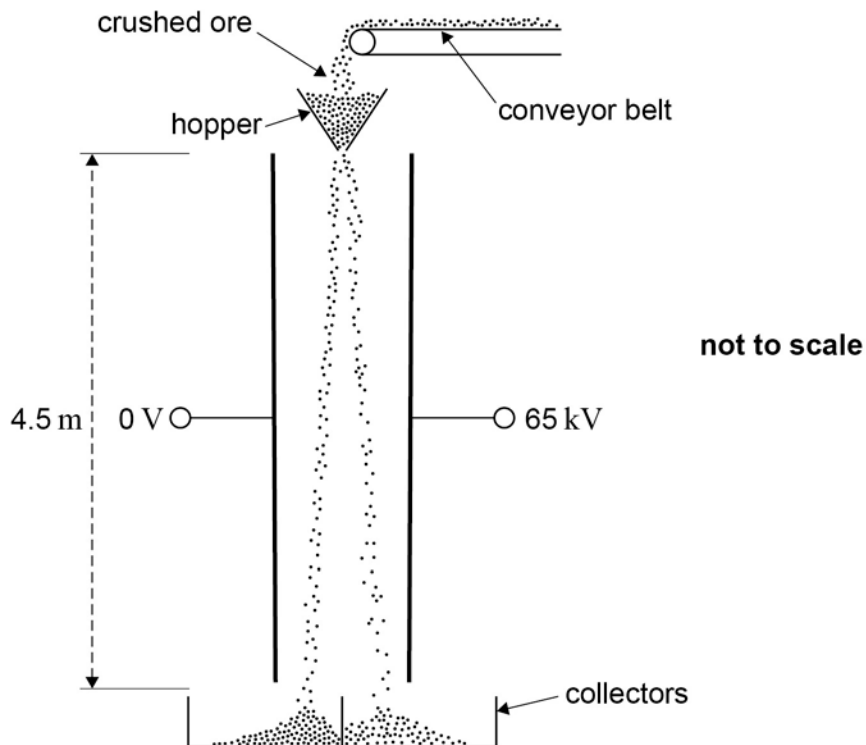
10



Do not write

Figure 1 shows a system that separates two minerals from the ore containing them using an electric field.

Figure 1



The crushed particles of the two different minerals gain opposite charges due to friction as they travel along the conveyor belt and through the hopper. When they leave the hopper they fall 4.5 m between two parallel plates that are separated by 0.35 m.

Assume that a particle has zero velocity when it leaves the hopper and enters the region between the plates.

Calculate the time taken for this particle to fall between the plates.

[2 marks]

time taken = _____ s

0 3 . 2 A potential difference (pd) of 65 kV is applied between the plates.

Show that when a particle of specific charge $1.2 \times 10^{-6} \text{ C kg}^{-1}$ is between the plates its horizontal acceleration is about 0.2 m s^{-2} .

[3 marks]

0 3 . 3 Calculate the total horizontal deflection of the particle that occurs when falling between the plates.

[1 mark]

horizontal deflection = _____m

0 3 . 4 Explain why the time to fall vertically between the plates is independent of the mass of a particle.

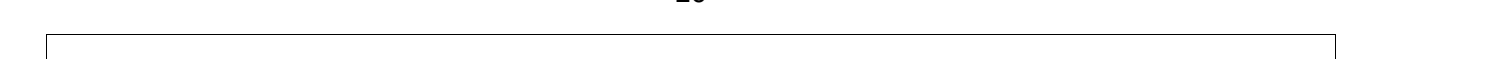
[2 marks]

0 3 . 5

State and explain **two** reasons, why the horizontal acceleration of a particle is different for each particle.

[3 marks]

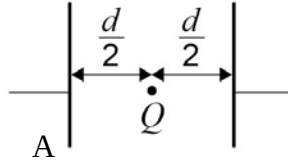
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Two parallel metal plates are separated by a distance d and have a potential difference V across them.

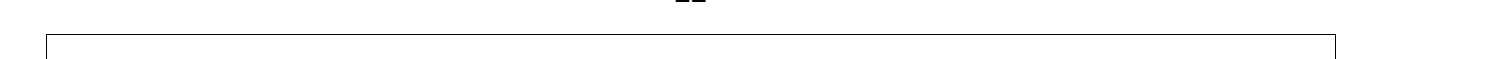
Which expression gives the magnitude of the electrostatic force acting on a charge Q placed midway between the plates?

[1 mark]



- A $\frac{2VQ}{d}$ ☐
- B $\frac{VQ}{d}$ ☐
- C $\frac{VQ}{2d}$ ☐
- D $\frac{Qd}{V}$ ☐

Turn over for the next question



1**2**

The electric potential at a distance r from a positive point charge is 45 V. The potential increases to 50 V when the distance from the point charge decreases by 1.5 m. What is the value of r ?

[1 mark]**A**

1.3 m

☐**B**

1.5 m

☐**C**

7.9 m

☐**D**

15 m

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